

1. Write the function of area of rectangle

```
def rectArea(b: Int, w: Int): Int = (w * b)
def rectArea(b: Int, w: Int): Int

rectArea(4,7)
val res60: Int = 28
```

2. Write the function to check whether a number is Prime or not

```
def checkPrime(value: Int): Boolean = {res = true;
  | for(i <- 2 to value-1)
  | if (value % i == 0)
  | res = false;
  | res;
  | }
def checkPrime(value: Int): Boolean
```

3. Write a Scala program which produces all values from an Array after removing duplicates

```
var numbers = Array(1,2,3,1,2,3,4,5)
var numbers: Array[Int] = Array(1, 2, 3, 1, 2, 3, 4, 5)

var unique = numbers.distinct
var unique: Array[Int] = Array(1, 2, 3, 4, 5)
```

4. Create a Map of prices for a number of shopping items you want. Now create another map from it with the same key, but prices slashed by 10%.

```
val prices = Map("Water" -> 68,
  | "Eggs" -> 69,
  | "Twix" -> 100,
  | "Coconut Water" -> 50)
val prices: scala.collection.immutable.Map[String,Int] = Map(Water -> 68,
Eggs -> 69, Twix -> 100, Coconut Water -> 50)

val discounted = prices.map{ case (item, price) => (item, price *0.9) }
val discounted: scala.collection.immutable.Map[String,Double] = Map(Water
-> 61.2, Eggs -> 62.1, Twix -> 90.0, Coconut Water -> 45.0)
```

5. Write a function minmax(values: Array[Int]) which returns a pair of minimum and maximum values of an array passed to it

```
def minmax(values: Array[Int]): (Int, Int) = {
  | var min = values(0); var max = values(0);
```

```

    | for(v <- values){
    | if (v < min) min = v;
    | if (v > max) max = v;
    | }
    | (min, max)
    | }
def minmax(values: Array[Int]): (Int, Int)

```

numbers

```
val res0: Array[Int] = Array(1, 2, 3, 1, 2, 3, 4, 5)
```

minmax(numbers)

```
val res1: (Int, Int) = (1,5)
```

6. Repeat above exercise as a procedure which prints the min and max values of an Array

```

def minmaxProc(values: Array[Int]): Unit = {
    | var min = values(0);
    | var max = values(0);
    | for(v <- values){
    | if(v < min) min = v;
    | if(v > max) max = v;
    | }
    | println("Min is: "+min);
    | println("Max is: "+max);
    | }
def minmaxProc(values: Array[Int]): Unit

```

minmaxProc(numbers)

```
Min is: 1
```

```
Max is: 5
```